

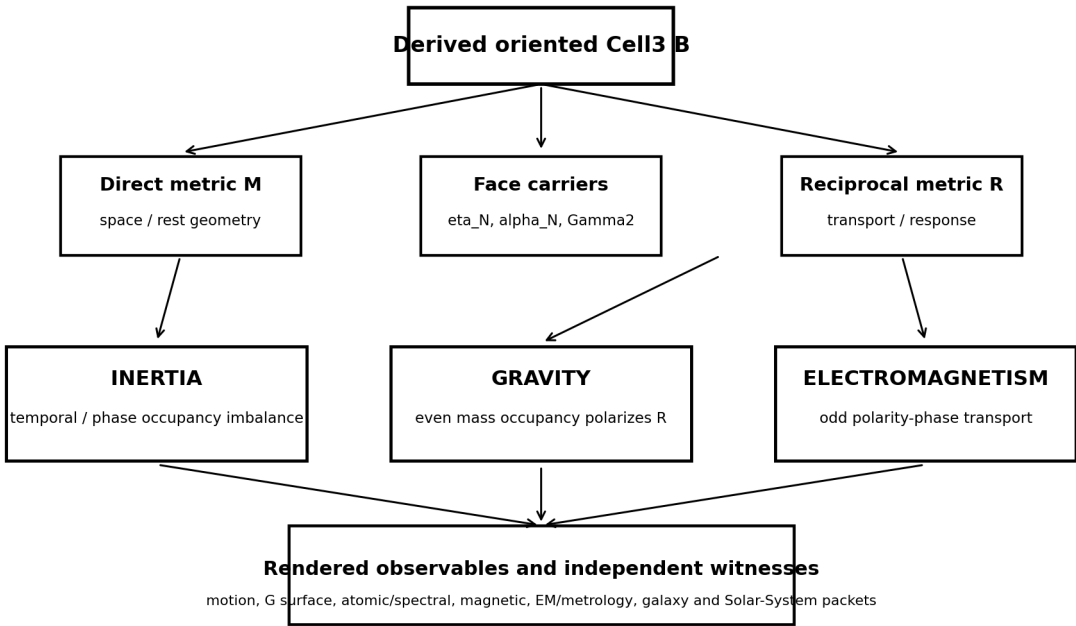
K-Field Converged Closure

Strict Geometry-First Derivation of Gravity, Inertia, and Electromagnetism
from the Derived Cell3 Geometry

Principal Investigator: P. Dwayne Esterline
Krytronx

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Converged K-Field closure: one geometry, three typed branch readings



Title figure. The converged closure begins with the derived oriented Cell3 geometry. Its direct metric, face-carrier triangle, reciprocal metric, native unit grammar, and branch compilers yield three typed physical readings: inertia as temporal-phase occupancy imbalance, gravity as even Mass-Bridge occupancy polarizing reciprocal geometry, and electromagnetism as odd polarity-phase transport returning to charge, action, fine/speed, magnetic, and metrology closure.

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Abstract

This reference presents the converged K-Field closure as a strict dependency graph beginning with the derived, ordered, ICRS-oriented Cell3 basis. The construction does not begin from independent force laws. It begins from one oblique metric cell whose primitive vectors generate a direct metric, reciprocal metric, three primitive faces, a centered three-state address grammar, alternating parity, seven-sheet organization, a three-channel phase-transfer operator, and a closed carrier triangle.

The three Cell3 face areas generate the native Mass Bridge η_N , the fine/speed pair α_N and c_N , and the high-order gravity scalar Γ_2 . The closure graph then supplies native update time, native causal rate, mass, action, and charge conventions before dimensional rendering. The full-cell closure length L_G yields a light-crossing clock T_G and its electron-Compton ratio S_{CMB} . The fine-leg bridge yields v_F , L_N , τ_N , and the dimensional gravity conversion $C_{G,pred}$. No additional native time coordinate or propagation speed is required.

The reciprocal metric supplies the native transport distance and the source-loaded operator $L_K = -\text{div}_K(\text{Rhat grad}_K)$. Native mass occupancy N_m becomes Mass-Bridge polarization charge $Q_m = \eta_N N_m$, producing an anisotropic inverse-distance potential and inverse-square response. Γ_2 and the dimensional bridge generate G_{base} ; the directional compiler generates G_{eff} . Inertia is the temporal, phase-count, and energy mismatch of matter traversing the fixed Cell3 closure length. Electromagnetism is the odd parity-phase branch generated from address polarity, native charge and action, three-channel transport, the fine/speed pair, electromagnetic-medium relations, quantum electrical metrology, and the magneton return to η_N .

Principal result. The least complete closure is: derived Cell3 geometry $\rightarrow$ primitive faces and reciprocal geometry $\rightarrow$ carrier triangle $\rightarrow$ native unit grammar $\rightarrow$ dimensional clocks and rates $\rightarrow$ reciprocal transport $\rightarrow$ three typed branch readings. Gravity is the even occupancy branch, inertia is the rate-of-occupancy branch, and electromagnetism is the odd polarity-phase branch. All three return to the same ordered Cell3 geometry.

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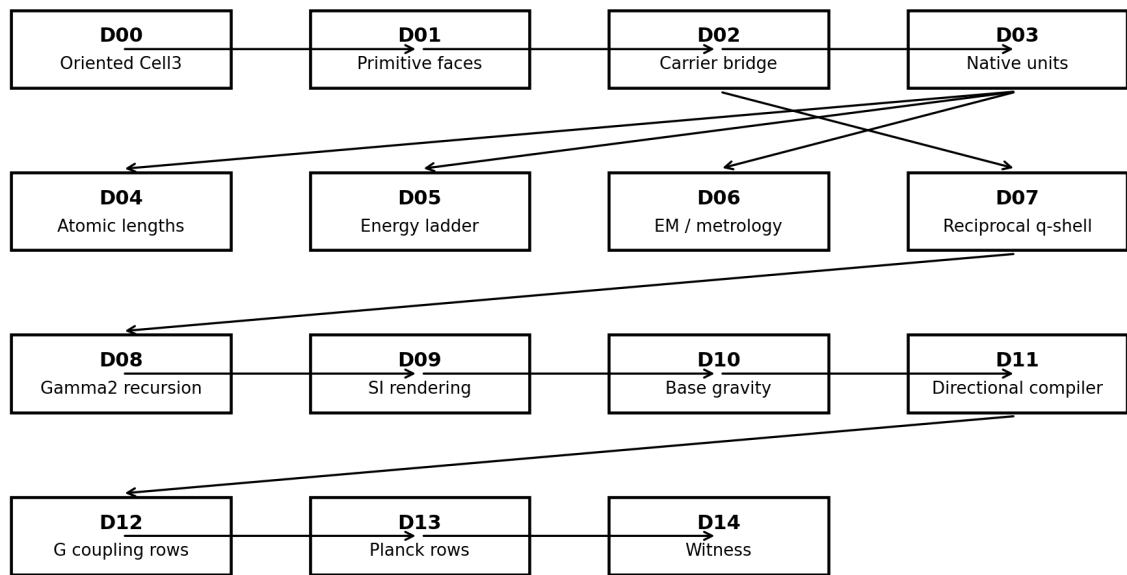
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1. Logical contract and dependency order

The closure is evaluated as a directed typed graph. Every result must be downstream of the derived Cell3 basis or of a declared rendering input. A witness value may test a completed branch but may not serve as an upstream cause. The strict order is encoded by the D00-D14 architecture.



Strict dependency order: geometry -> native closure -> branch return -> witness

Figure 1. D00-D14 dependency architecture. Geometry and native ratios are generated first. Atomic, energy, electromagnetic, reciprocal, gravity, and observer branches are then derived. Witness comparisons occur only after the model quantities exist.

Node	Object	Logical role
D00	Base oriented Cell3 geometry	Generator root
D01	Primitive face-area layer	Native geometric generator
D02	Face-channel carrier bridge	Mass, fine/speed, and cross-leg ratios
D03	Native unit definitions	Closure time, causal rate, action, charge, and mass conventions
D04	Atomic lengths and scattering	Derived native output
D05	Energy ladder	Derived native output
D06	Electromagnetic and metrology skeleton	Derived native branch return
D07	Reciprocal geometry and q-shell standards	Native transport geometry
D08	Gamma2 recursion	Native gravity scalar
D09	Dimensional rendering bridge	Permitted conversion layer
D10	Base gravity construction	Gamma2 multiplied by the dimensional conversion
D11	Observer-indexed directional layer	Directional compiler
D12	G-derived couplings	Downstream gravity family
D13	Planck-family rows	Downstream combined family
D14	Witness comparison	Terminal validation layer

The closure may therefore be read in four bands: geometry, native closure, rendering, and branch return. This prevents a downstream physical label from being mistaken for a primitive cause.

2. Authoritative Cell3 geometry

The root object is the ordered basis $B=[a \ b \ c]$. Its columns are physical ICRS vectors in nanometres. Ordering is structural: it fixes branch identity, handedness, face assignment, reciprocal directions, parity conventions, and all later response matrices.

Authoritative Cell3: eight vertices, twelve edges, three origin faces, and sheet normal

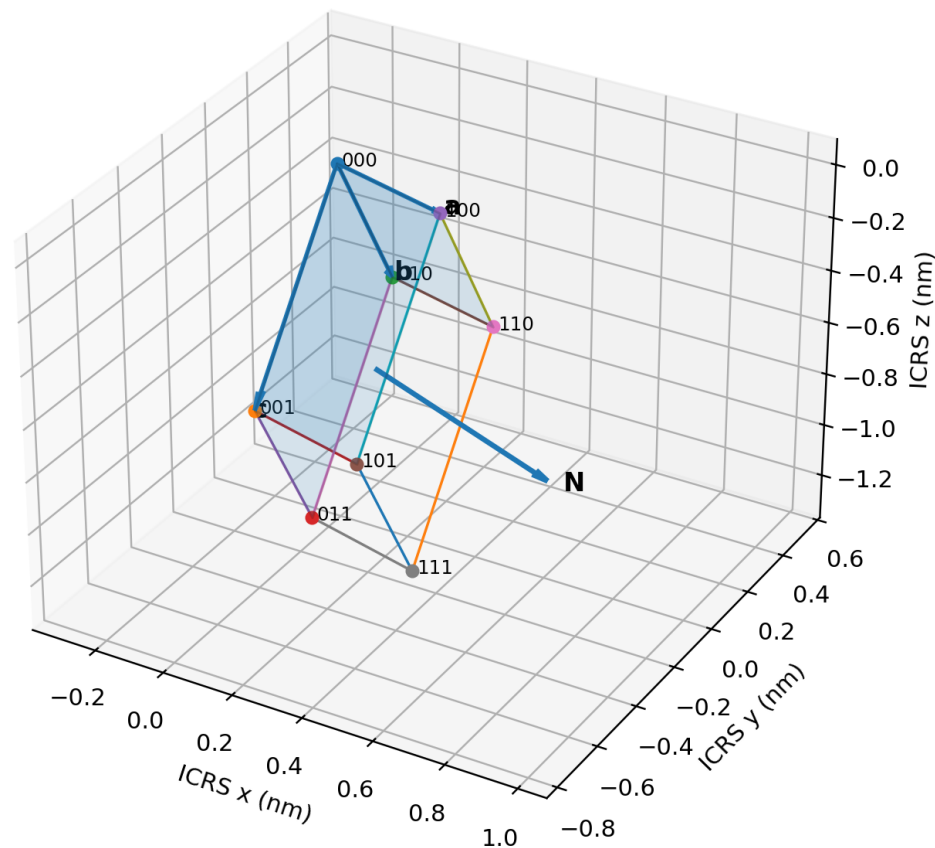


Figure 2. Dimensionally correct Cell3 in ICRS nanometres. All eight vertices, twelve primitive edges, three origin faces, the ordered vectors a , b , c , and the metric-dual sheet normal N are computed from the authoritative basis.

Object	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	Norm (nm)
a	0.327501242069651	-0.054897007886313	-0.037487502896305	0.334179679073152
b	0.035330775844051	0.207319850441498	-0.547401085391957	0.586410890414836
c	-0.010129317626665	-0.389787702363804	-0.689734408362922	0.792319765037694
N	0.876976866366201	0.047223124229988	-0.478206600118064	1.000000000000000

2.1 Direct metric, volume, and closure length

$M = B^T B$ $R = M^{-1}$ $V = \det B $ $L_G = a+b+c $		
Quantity	Value	Unit / meaning
$\det(B)$	-0.117914891916699	nm^3 ; ordered handedness is negative
Cell volume	0.117914891916699	nm^3
Closure length L_G	1.343653621680000	nm

Quantity	Value	Unit / meaning
Half-step sheet spacing h	0.151272688042485	nm
Native-to-nm edge scale	16.068602900842745	nm per native length unit

The closure length L_G is the physical body diagonal. It is not an arbitrary reference scale; it is generated by the ordered cell itself.

2.2 Primitive faces and plane pairs

Three Cell3 face-plane pairs from exact cross products

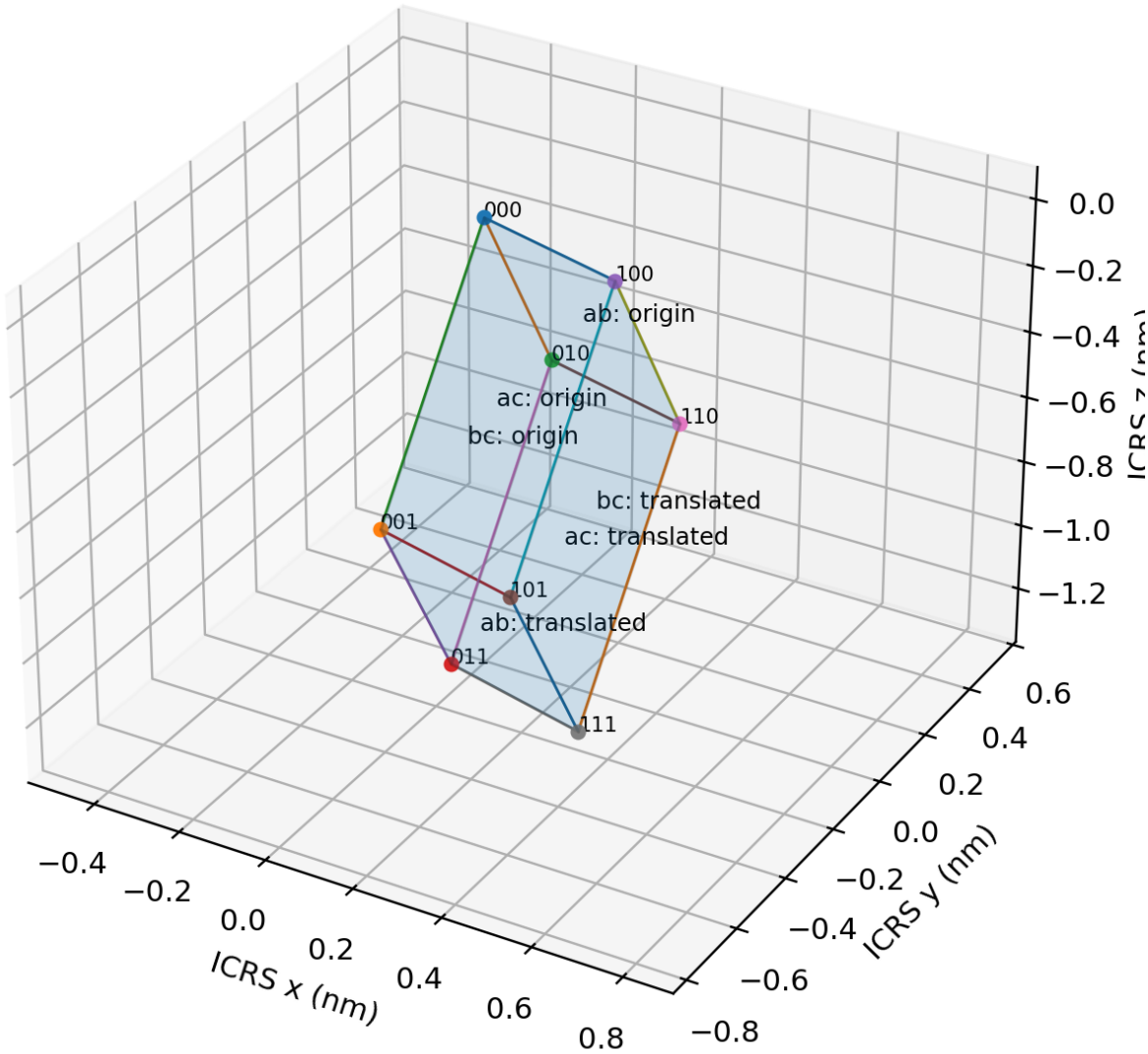


Figure 3. The three origin faces and their translated opposite faces. Each plane is determined by an exact cross product and one translated offset. No independent plane parameter is introduced.

Face	Physical area (nm^2)	Unit normal	Opposite-plane offset (nm)
ab	0.194869165469900	(0.194092201, 0.913177169, 0.358379235)	-0.605097741514743
ac	0.261106235799531	(0.089052594, 0.866576769, -0.491033948)	0.451597379724130
bc	0.357809539128562	(-0.995965190, 0.083602163, -0.032619291)	-0.329546529709293

The physical face areas are rendered in nm^2. The carrier formulas use the corresponding native-unit face areas. Because the cell is uniformly rendered, both representations preserve all ratios exactly.

3. Hidden three-state grammar, Lucas window, and parity

The exact hidden coordinate rule inside Cell3 is $\{0,1/2,1\}^3$. Doubling produces $\{0,1,2\}^3$. Re-centering about the body center produces the Lucas alphabet $\{-1,0,1\}^3$. The finite 27-node body-centred neighborhood is therefore not an added lattice: it is the exact centered completion of the base cell.

$$p(i,j,k) = B[i,j,k]^T / 2$$
$$\lambda(i,j,k) = i+j+k$$
$$N \cdot p(i,j,k) = h \lambda$$
$$\chi(i,j,k) = (-1)^{(i+j+k)}$$

Complete 27-node Lucas window on seven metric-dual sheets

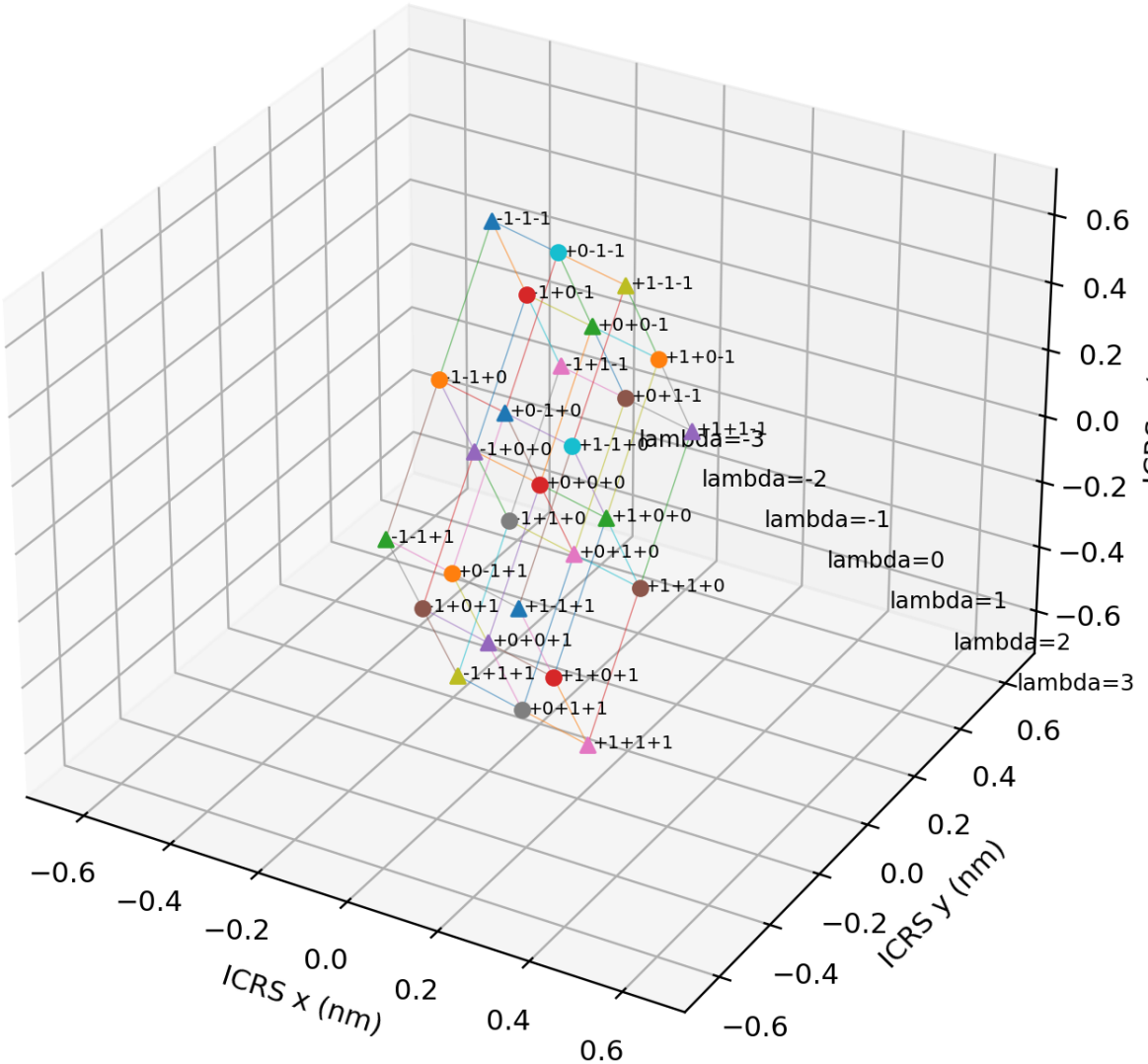


Figure 4. Complete 27-node Lucas window. Every integer address in $\{-1,0,1\}^3$ is plotted at its dimensionally correct ICRS coordinate. The seven sheet populations are 1, 3, 6, 7, 6, 3, 1. Circle and triangle markers distinguish even and odd parity.

Sheet λ	Population	Offset along N (nm)	Polarity
-3	1	-0.453818064127454	negative
-2	3	-0.302545376084970	positive

Sheet lambda	Population	Offset along N (nm)	Polarity
-1	6	-0.151272688042485	negative
0	7	0.000000000000000	positive
1	6	0.151272688042485	negative
2	3	0.302545376084970	positive
3	1	0.453818064127454	negative

The address sum is simultaneously a sheet coordinate and a parity coordinate. Its kernel contains the same-sheet increments and forms the rank-two A2 root lattice. The three primitive lifts therefore share one normal advance while retaining distinct in-sheet registry slips.

4. Carrier triangle and native constant generation

The three primitive faces are one closed carrier triangle, not three adjustable constants. Their native-unit areas are derived by dividing the dimensioned basis by the single native-to-nm scale. The ratios are independent of that scale.

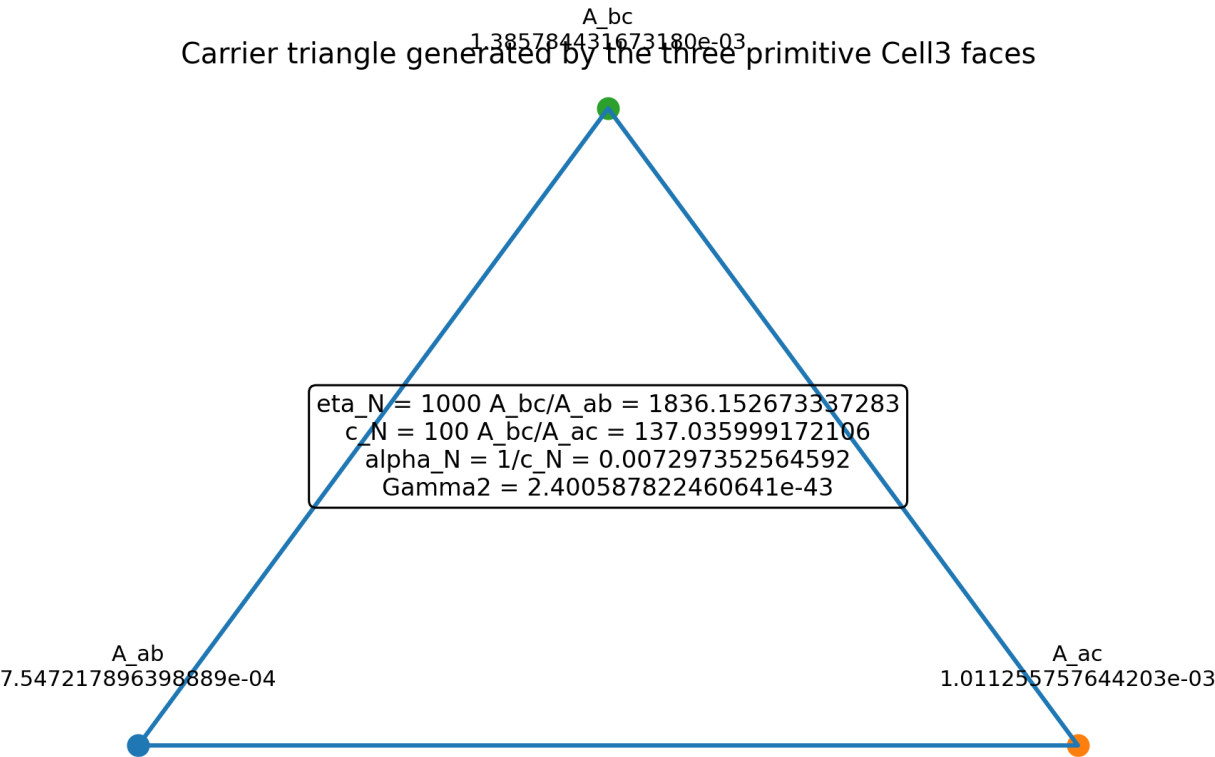


Figure 5. Carrier triangle. The face areas generate the Mass Bridge η_N , the inverse fine/speed factor c_N , the fine factor α_N , and the high-order gravity scalar Γ_2 .

$\begin{aligned}\eta_N &= 1000 A_{bc}/A_{ab} \\ c_N &= 100 A_{bc}/A_{ac} \\ \alpha_N &= 1/c_N \\ \mu_N &= \eta_N/(1+\eta_N) \\ \Gamma_2 &= A_{ab}^9 A_{ac}^8 / (100^{11} A_{bc}^{11})\end{aligned}$		
Quantity	Expression	Current value
A_ab	a_native x b_native	7.54721789639889306e-04
A_ac	a_native x c_native	1.011255757644202984e-03
A_bc	b_native x c_native	1.385784431673180205e-03
eta_N	1000 A_bc/A_ab	1836.152673337282522
c_N	100 A_bc/A_ac	137.035999172105591

Quantity	Expression	Current value
alpha_N	1/c_N	7.297352564592058874e-03
mu_N	eta_N/(1+eta_N)	0.999455679424735
Gamma2	A_ab^9 A_ac^8 /(100^11 A_bc^11)	2.400587822460640964e-43

The carrier triangle is the common hinge of the three physical branches. Eta_N supplies the mass hierarchy, alpha_N and c_N supply the fine/speed hierarchy, and Gamma2 supplies the native gravitational scalar. Later magnetic and electromagnetic quantities return to the same sources.

5. Native units, time, velocity, and dimensional rendering

The closure graph already contains native time and native propagation. One reproduction or update of the complete closure object is one native time unit. One primitive displacement per update is the native causal rate.

```

Rep_R[U_R(k)] = U_R(k+1)
tau_R = 1
t_R = N tau_R = N
c_R = max(Delta s_R/Delta N) = 1 for a primitive unit transition
    
```

Closed native-time hierarchy and fine-leg dimensional bridge

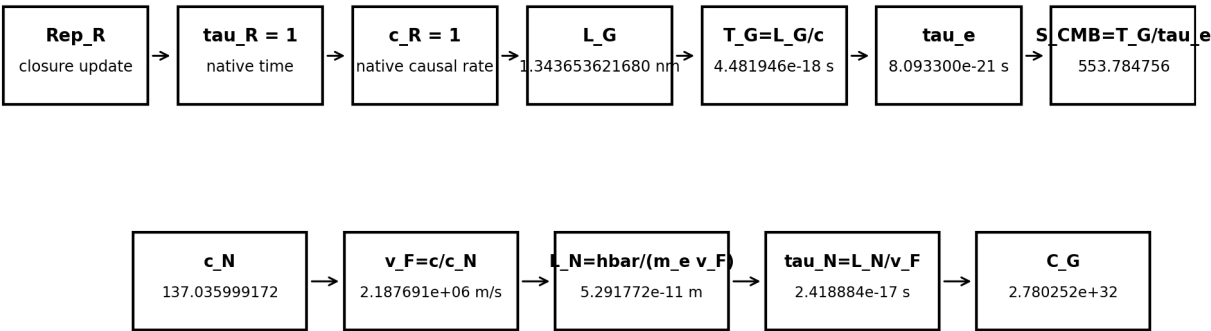


Figure 6. Closed temporal hierarchy. The upper row shows native update time and the full-cell traversal ruler. The lower row shows the fine-leg speed, length, time, and gravity conversion.

5.1 Full-cell temporal ruler

```

L_G = |a+b+c|
T_G = L_G/c
tau_e = h/(m_e c^2)
S_CMB = T_G/tau_e
T(beta) = T_G/beta
    
```

Quantity	Value	Role
L_G	1.343653621680000 nm	Fixed Cell3 closure length
T_G	4.481946045754091864e-18 s	Light-crossing time of L_G
tau_e	8.093299783346508991e-21 s	Electron Compton period
S_CMB	553.784755999838	Closure-time ratio T_G/tau_e

Quantity	Value	Role
tau_R	1 native update	Closure-native time unit
c_R	1 native length per update	Closure-native causal rate

5.2 Fine-leg dimensional bridge

$$\begin{aligned} v_F &= c/c_N \\ L_N &= \hbar/(m_e v_F) \\ \tau_N &= L_N/v_F \\ C_{G,pred} &= L_N^3/(m_e \tau_N^2) \\ G_{base} &= \Gamma^2 C_{G,pred} \end{aligned}$$

Quantity	Value	Unit / role
v_F	2.187691262231657e+06	m/s
L_N	5.291772105255611e-11	m
tau_N	2.418884326418843e-17	s
C_G,pred	2.780252249786785e+32	m^3 kg^-1 s^-2 per Gamma2
Gamma2	2.400587822460641e-43	native gravity scalar
G_base	6.674239694206957e-11	m^3 kg^-1 s^-2
Versioned packet G_base	6.674239691094150e-11	directional packet replay value
Corpus G witness	6.67430000e-11	terminal comparison value

The two rendered times have distinct roles. T_G is the full Cell3 light traversal. Tau_N is the fine-leg time used by the atomic and gravity conversion branch. Both are descendants of the native rule time = path length / path rate.

6. Reciprocal geometry and the transport operator

The direct metric M describes physical separation in the ordered Cell3 basis. Its inverse describes reciprocal sensitivity. Dividing the reciprocal metric by its mean eigenvalue scale produces the normalized response metric Rhat.

$$\begin{aligned} \text{qbar}_F &= \text{trace}(M^{-1})/3 \\ \text{Rhat} &= M^{-1}/\text{qbar}_F \\ r_K^2 &= x_K^T \text{Rhat}^{-1} x_K \\ L_K &= -\text{div}_K(\text{Rhat grad}_K) \end{aligned}$$

Rhat row	a	b	c
a	1.640131844067	0.000273378731	-0.114920776385
b	0.000273378731	0.873392421764	-0.412378656942
c	-0.114920776385	-0.412378656942	0.486475734169
Invariant		Value	
det(Rhat)		0.406442789872079	
sqrt(det(Rhat))		0.637528658079053	
qbar_F		5.614198371912021 nm^-2	

Unit reciprocal-metric distance surface: $x^T \hat{R}^{-1} x = 1$

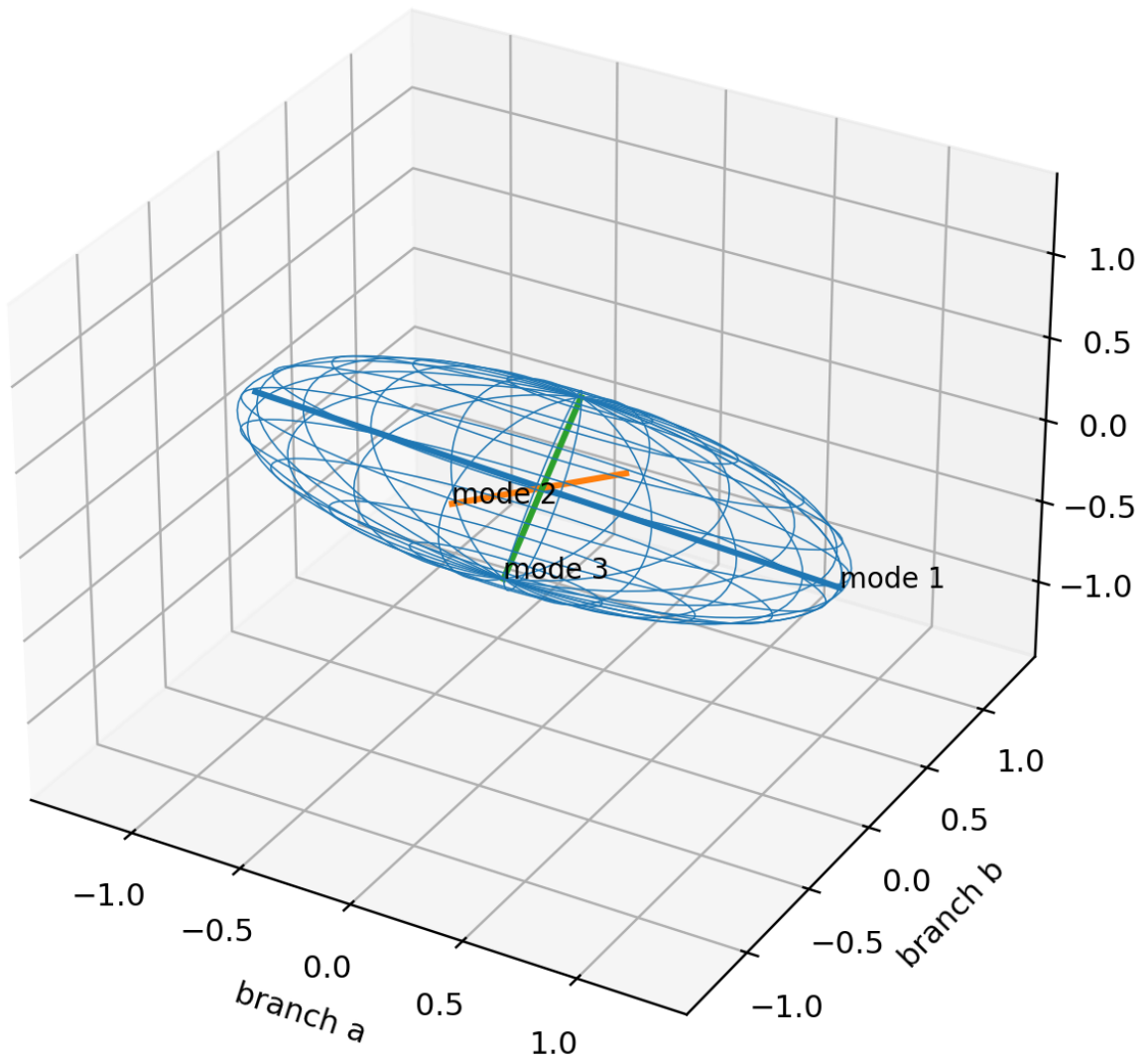


Figure 7. Unit reciprocal-metric distance surface. The ellipsoid is generated directly from $\hat{R}$ and visualizes the anisotropic native distance used by the source-loaded closure field.

Point-source potential contours from the normalized reciprocal metric

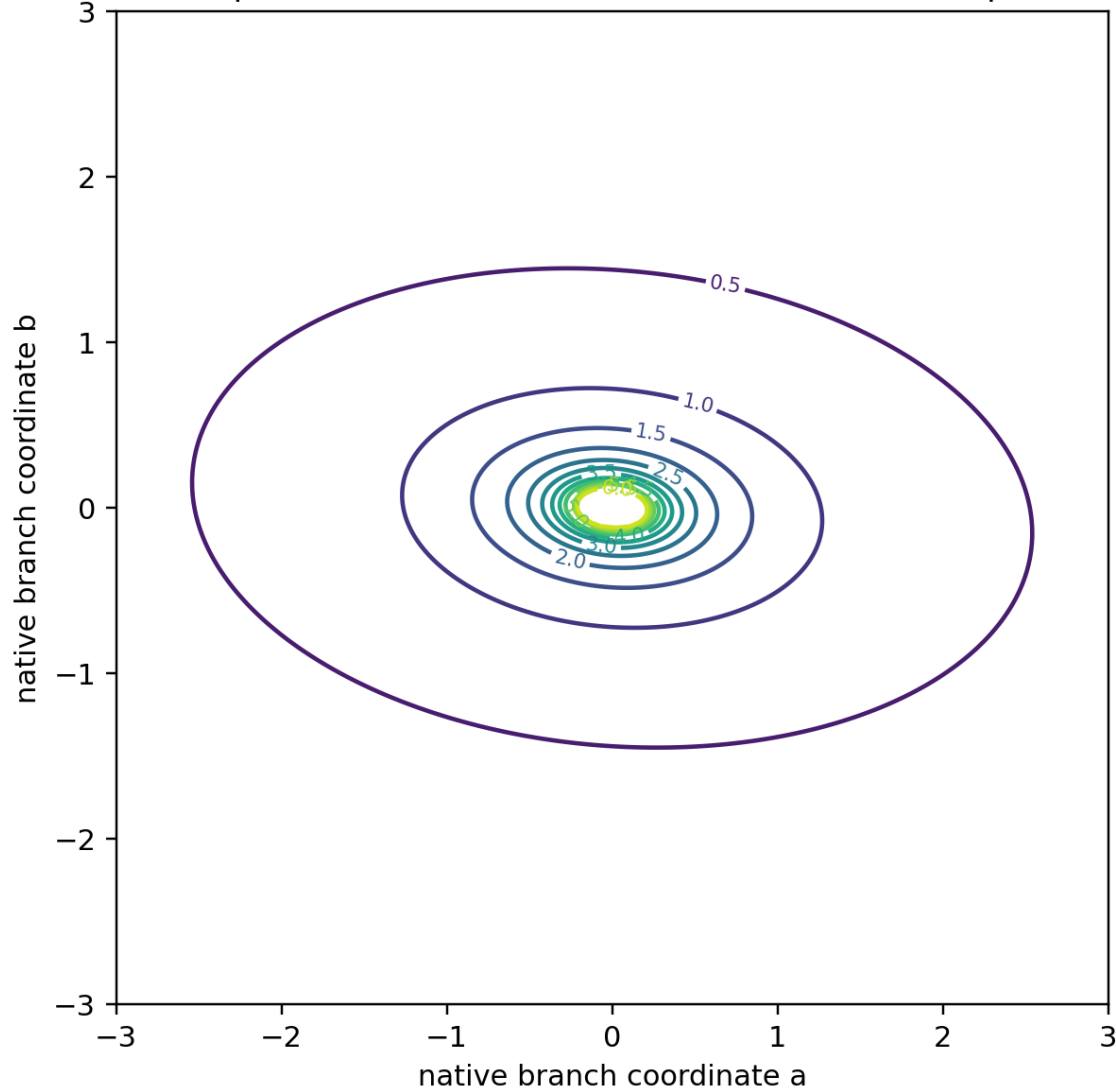


Figure 8. Point-source potential contours in a branch-coordinate plane. The contours are level sets of $1/r_K$, where r_K is defined by the normalized reciprocal metric.

The reciprocal metric is therefore the native geometry of propagation, convergence, and directional sensitivity. It is not a separately selected inertia or gravity tensor.

7. Three-channel laminar phase transport

A forward sheet transition can occur along branch a, b, or c. The three routes advance the same sheet coordinate but retain distinct in-sheet slips. The complete local transfer is the coherent sum of their three phases.

```

Psi_(lambda+1)(k) = -H(k) Psi_lambda(k)
H(k) = [exp(-i k.t_a)+exp(-i k.t_b)+exp(-i k.t_c)]/3
psi_ac = psi_ab + psi_bc (mod 2 pi)

```

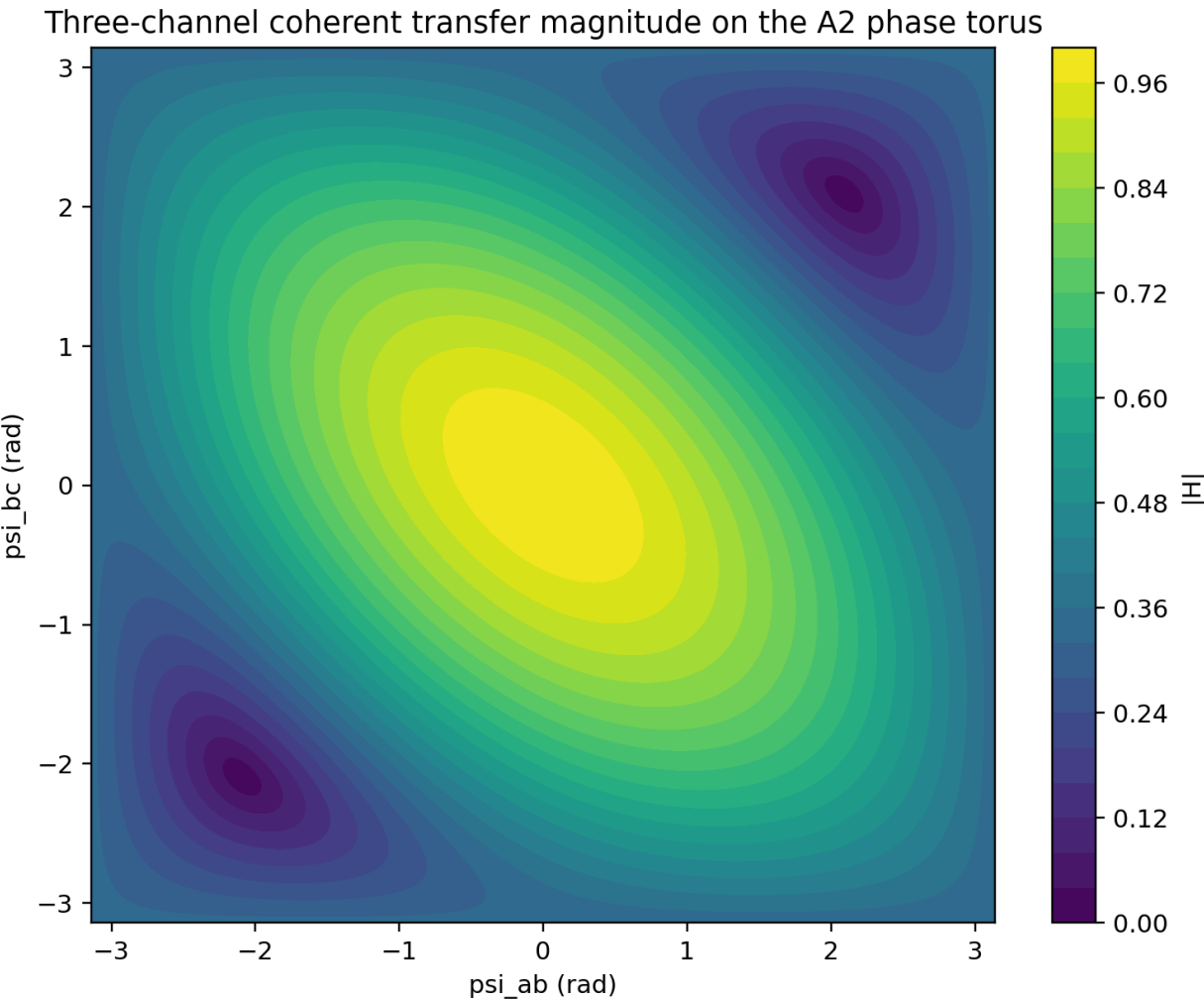


Figure 9. Magnitude of the three-channel coherent transfer on the two-dimensional A2 phase torus. Reinforcement and cancellation are exact consequences of the three-route closure.

7.1 Weighted registry-frustration energy

$$D = w_{ab}[1-\cos(\psi_{ab})] + w_{bc}[1-\cos(\psi_{bc})] + w_{ac}[1-\cos(\psi_{ab}+\psi_{bc})]$$
$$D = (1/2) \mathbf{z}^* \mathbf{L}_w \mathbf{z}$$
$$\mathbf{L}_w = \mathbf{C}^T \mathbf{W} \mathbf{C}$$

Channel	Weight	Share of total
ab	0.396285985688453	0.051818%
ac	166.587550282441185	21.782922%
bc	597.778333979951640	78.165259%

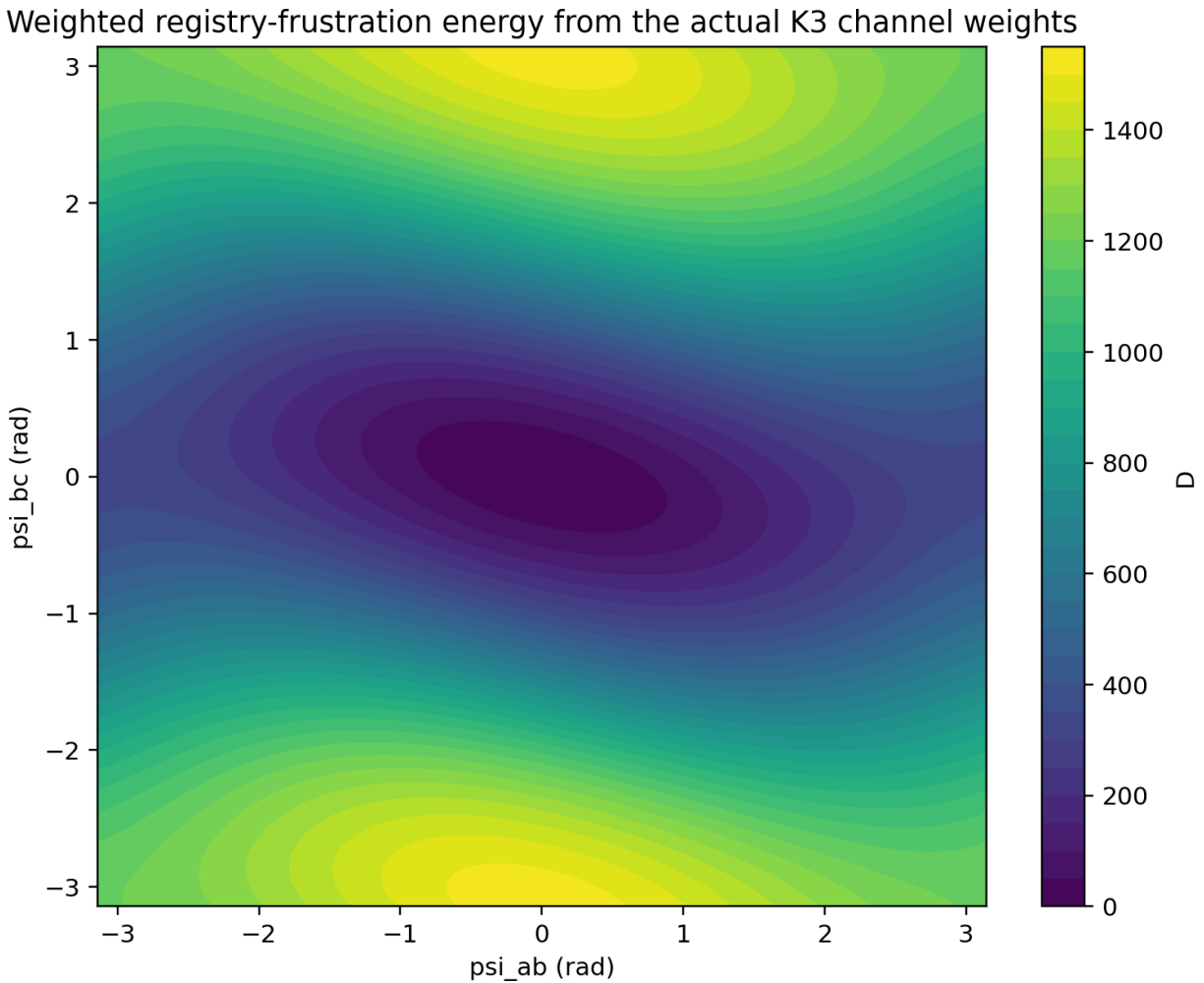


Figure 10. Weighted phase-frustration energy from the actual channel weights. The bc exchange channel dominates, the ac channel is secondary, and the ab channel is fine.

The sheet-normal sign reversal and the in-sheet phase shifts belong to one transfer object. This is the common phase mechanism later read as registry response, directional response, and odd electromagnetic transport.

8. Gravity: even Mass-Bridge occupancy polarizing reciprocal geometry

Gravity begins with occupancy, not with a primitive force constant. A native mass occupancy N_m is converted by the Mass Bridge into the source charge Q_m . The source is then propagated through the already-derived reciprocal metric.

```
Q_m = sigma_m N_m
sigma_m = eta_N
L_K Phi_m = sigma_m rho_m
-div_K(Rhat grad_K Phi_m) = Q_m(x_K)
```

8.1 Point-source solution

```
Q_m(x_K) = sigma_m N_m delta_K(x_K)
Phi_m(x_K) = [sigma_m N_m / (4 pi sqrt(det Rhat))] (1/r_K)
r_K^2 = x_K^T Rhat^-1 x_K
g_K = -grad_K Phi_m
```

The inverse-square form follows by differentiating the inverse reciprocal distance. The anisotropy remains entirely in Rhat and in the directional compiler; no second spatial metric is introduced.

8.2 Base gravity and directional rendering

```
G_base = Gamma2 C_G,pred
f_dir(n) = ||A_cell3 n||
Phi_G(0,n) = f_obs/f_dir(n)
G_eff(n) = G_base Phi_G(0,n)
G_phase = A_cell3^T A_cell3
```

Object	Value / expression	Role
Gamma2	2.400587822460641e-43	Native scalar seed
C_G,pred	2.780252249786785e+32	Dimensional conversion
G_base	6.674239694206957e-11	Pre-directional scalar gravity
f_obs	1.447739	Observer baseline response
f_dir	$\sqrt{n^T G_phase\ n}$	Direction-dependent response
G_eff	$G_base\ f_obs/f_dir$	Observer-direction rendered gravity

The directional response is antipodally even because it depends on a quadratic form. Its extrema are the eigenvectors of G_phase . The quadratic surface is the low-order directional envelope of the deeper phase-holonomy response.

9. Inertia: temporal and phase-occupation imbalance

In the locked-cell formulation, acceleration does not require Cell3 to deform. The rest geometry and the face-channel Mass Bridge remain fixed. Motion changes the time spent in the closure length, the number of Compton clock periods, the de Broglie phase count, and the energy carried by the transit clock.

```
T(beta) = T_G/beta
N_C(beta) = T(beta)/tau_e = S_CMB/beta
N_dB(beta) = gamma beta S_CMB
E_transit(beta) = h/T(beta) = beta h c/L_G
K(beta) = (gamma-1) m c^2
Delta E_I(beta) = K(beta) - E_transit(beta)
```

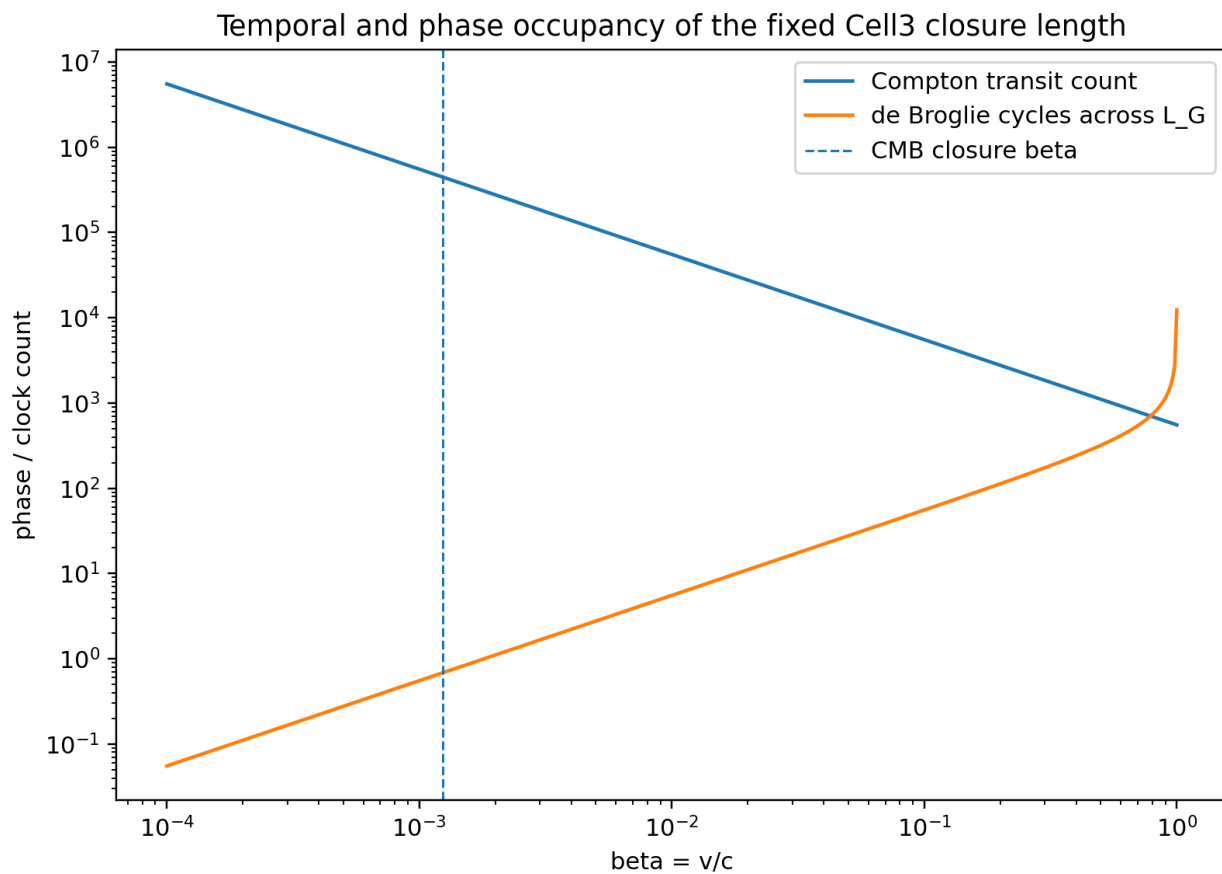



Figure 11. Compton transit count and de Broglie phase count across the fixed Cell3 closure length. The two counts move oppositely as beta changes.

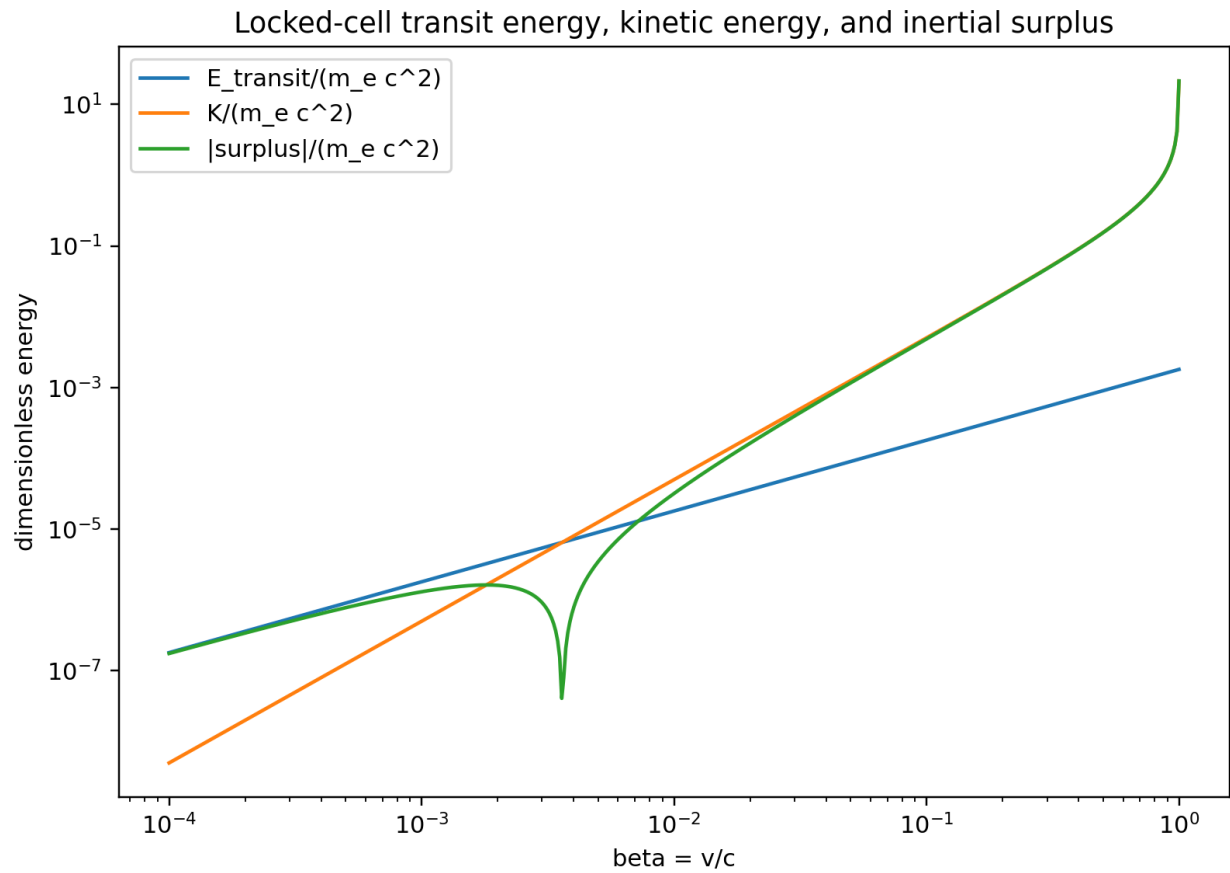


Figure 12. Transit-clock energy, relativistic kinetic energy, and their surplus. The transit-clock branch is linear in β , while kinetic energy becomes nonlinear.

β	T/T_G	N_{C/S_CMB}	N_{dB/S_CMB}	$E_{\text{trans}}/(m_e c^2)$	$K/(m_e c^2)$
0.001233562	810.660693090	810.660693090	0.001233563	2.227511155e-06	7.608381285e-07
0.010000000	100.000000000	100.000000000	0.010000500	1.805755737e-05	5.000375031e-05
0.100000000	10.000000000	10.000000000	0.100503782	1.805755737e-04	5.037815259e-03
0.500000000	2.000000000	2.000000000	0.577350269	9.028778683e-04	1.547005384e-01
0.900000000	1.111111111	1.111111111	2.064741605	1.625180163e-03	1.294157339e+00
0.990000000	1.010101010	1.010101010	7.017923930	1.787698179e-03	6.088812050e+00

Inertia is therefore the cost of changing the temporal occupation of a fixed Cell3 closure. The effective low-speed mass response is a derived approximation to this clock, phase, and energy imbalance, not its primitive source.

10. Equivalence of gravitational and inertial source charge

The same Mass-Bridge factor that converts occupancy into gravitational charge also converts occupancy into rendered inertial mass. When both branches use the same source identity, the factor cancels from the ratio that defines the rendered gravitational constant.

Mass-Bridge source identity and equivalence cancellation

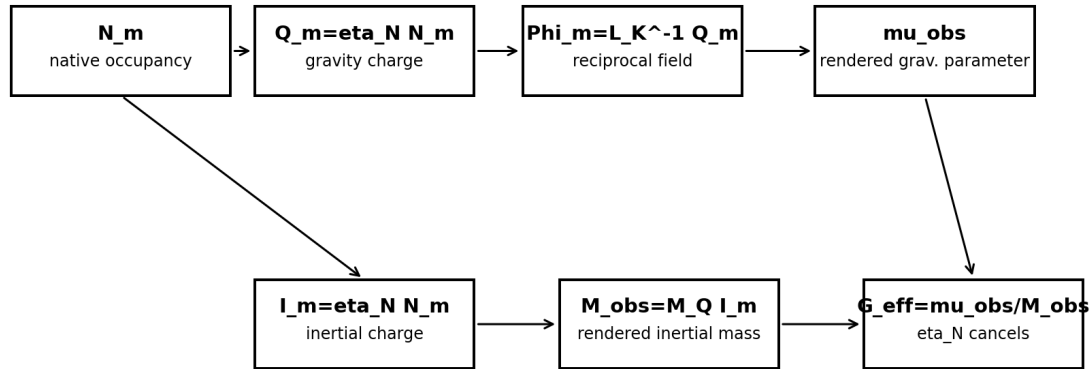


Figure 13. Equivalence cancellation. The gravitational and inertial branches share $Q_m = I_m = \eta_N N_m$. The source factor therefore cancels from $G_{\text{eff}} = \mu_{\text{obs}} / M_{\text{obs}}$.

```

Q_m = I_m = sigma_m N_m
M_obs = M_Q sigma_m N_m
mu_obs = C_mu N_m B_G(n,0)
G_eff = mu_obs/M_obs
  
```

Free-fall universality is thus a native charge-identity condition. Gravity measures how the shared occupancy charge polarizes reciprocal geometry. Inertia measures how the same occupancy resists changed temporal closure. Their equivalence is forced when they are two renderings of the same source.

11. Electromagnetism: odd polarity-phase transport

The electromagnetic branch is generated by the sign and phase information already present in the closure graph. Address parity supplies the odd source sign. The native charge magnitude and action supply the unit skeleton. The three-channel transfer supplies local phase transport. The fine/speed pair supplies the constitutive scaling.

```

chi(i,j,k) = (-1)^(i+j+k)
q_N = chi q0,N
q0,N = 1
H_N = 1
c_N = 1/alpha_N
  
```

Electromagnetic branch: odd polarity-phase transport returning to charge, action, and mass

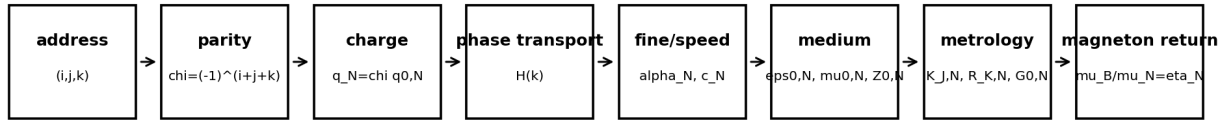


Figure 14. Electromagnetic closure. Address parity produces signed native charge; the three-channel operator transports phase; alpha_N and c_N generate the medium branch; action and charge generate the metrology branch; magnetons return to eta_N.

11.1 Native electromagnetic-medium relations

$$\begin{aligned}\epsilon_{0,N} &= 1/2 \\ \mu_{0,N} &= 2 \alpha_N^2 \\ Z_{0,N} &= 2 \alpha_N\end{aligned}$$

11.2 Charge-action metrology closure

$$\begin{aligned}K_{J,N} &= 2 q_{0,N}/H_N \\ R_{K,N} &= H_N/q_{0,N}^2 \\ G_{0,N} &= 2 q_{0,N}^2/H_N\end{aligned}$$

11.3 Magneton return to the Mass Bridge

$$\begin{aligned}\mu_{B,N} &= 1/2 \\ \mu_{N,N} &= 1/(2 \eta_N) \\ \mu_{B,N}/\mu_{N,N} &= \eta_N\end{aligned}$$

Branch object	Native expression	Dependency return
$\epsilon_{0,N}$	$1/2$	Charge-action-fine/speed skeleton
$\mu_{0,N}$	$2 \alpha_N^2$	Fine/speed pair
$Z_{0,N}$	$2 \alpha_N$	Fine/speed pair
$K_{J,N}$	$2 q_{0,N}/H_N$	Charge-action base
$R_{K,N}$	$H_N/q_{0,N}^2$	Charge-action base
$G_{0,N}$	$2 q_{0,N}^2/H_N$	Charge-action base
$\mu_{B,N}/\mu_{N,N}$	η_N	Mass Bridge

Electromagnetism is therefore the odd polarity-phase branch of the same closure state. Its medium, metrology, and magnetic quantities are not independent additions; they are return paths to alpha_N, c_N, q0,N, H_N, and eta_N.

12. Even, odd, and rate branch separation

The three physical readings are separated by transformation type rather than by introducing three unrelated primitive fields.

Typed reading	Native object	Transformation character	Physical interpretation
Even occupancy	N_m and Q_m	Invariant under polarity reversal	Mass and gravity
Odd polarity-phase	χ , q_N , $H(k)$	Changes sign or phase under conjugation	Charge and electromagnetism
Rate of occupancy	$T(\beta)$, N_C , N_{dB} , ΔE_I	Changes with traversal rate	Inertia

The common origin is the ordered Cell3 closure. Gravity is not a primitive force; it is the rendered even branch of mass occupancy polarizing reciprocal geometry. Electromagnetism is the rendered odd branch of polarity and phase transport. Inertia is the temporal derivative branch of occupancy through the fixed closure ruler.

13. Complete logical structured flow

Converged K-Field closure: one geometry, three typed branch readings

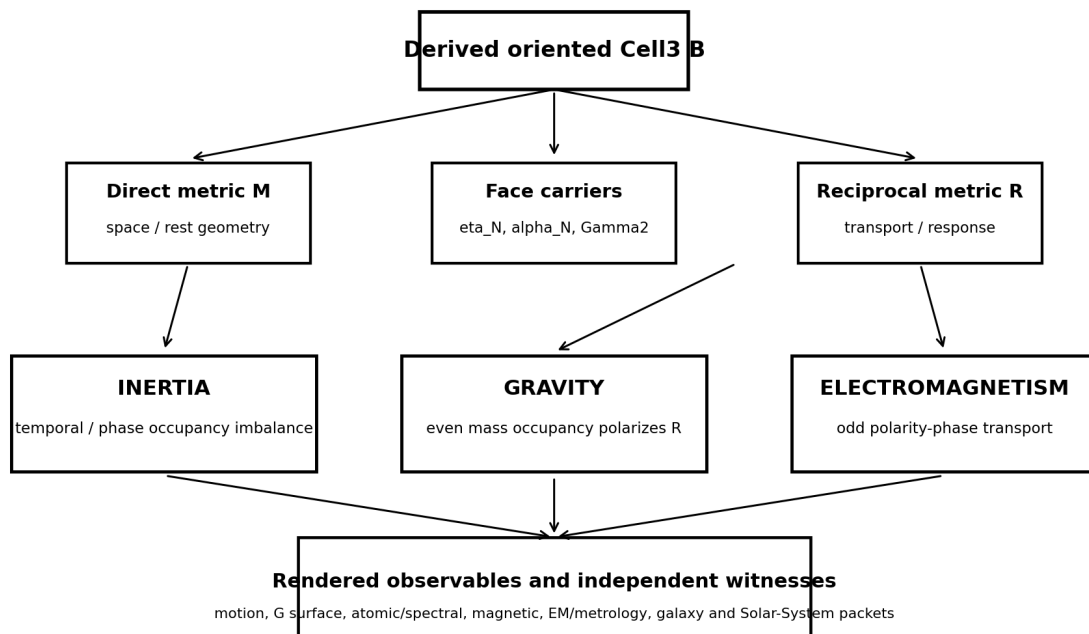


Figure 15. Final closure map. Cell3 supplies direct geometry, carrier geometry, and reciprocal geometry. The three branches are distinct readings of one closure state and terminate in rendered observables and independent witnesses.

13.1 Geometry and native closure

```

B -> {M,R,V,N,h,A_ab,A_ac,A_bc}
{A_ab,A_ac,A_bc} -> {eta_N,alpha_N,c_N,Gamma2}
Rep_R -> {tau_R,c_R,H_N,q0,N,m_e,N,m_p,N}

```

13.2 Dimensional rendering

```

B -> L_G -> T_G -> {tau_e,S_CMB,T(beta)}
c_N -> v_F -> L_N -> tau_N -> C_G,pred
Gamma2 C_G,pred -> G_base

```

13.3 Reciprocal transport

```
M^-1 -> Rhat -> {r_K, L_K}
{t_a, t_b, t_c, chi} -> H(k)
{s_ab, s_ac, s_bc, W} -> D and K_hol
```

13.4 Branch readings

```
Gravity: N_m -> Q_m -> Phi_m -> g_K -> G_eff
Inertia: beta -> T(beta) -> {N_C, N_dB, E_transit, Delta E_I}
Electromagnetism: chi -> q_N -> H(k) -> {medium, metrology, magneton return}
```

Every arrow is either a direct geometric calculation, an exact native definition, a declared dimensional rendering, or a terminal observer compiler. No independent force constant is inserted between Cell3 and the three branch readings.

14. Reconstruction algorithm

Step	Operation	Output
1	Enter the ordered ICRS basis B. Verify determinant, handedness, and body diagonal.	Authoritative Cell3
2	Compute M, R, V, N, h, and all primitive face areas and plane pairs.	Direct and reciprocal geometry
3	Generate {0,1/2,1} ³ , recenter to {-1,0,1} ³ , and compute lambda and chi.	Lucas window, sheets, parity
4	Compute native face areas and the carrier triangle.	eta_N, alpha_N, c_N, mu_N, Gamma2
5	Apply the native update and unit grammar.	tau_R, c_R, H_N, q0,N, mass references
6	Render L_G, T_G, tau_e, S_CMB, and the moving transit T(beta).	Full-cell temporal branch
7	Render v_F, L_N, tau_N, C_G,pred, and G_base.	Fine-leg and scalar gravity bridge
8	Normalize R to Rhat and construct r_K and L_K.	Native transport operator
9	Construct the three-channel transfer H(k) and weighted phase energy D.	Phase-holonomy response
10	Load occupancy N_m through Q_m=eta_N N_m and solve L_K Phi_m=Q_m.	Native gravity field
11	Apply the directional compiler to G_base.	G_eff(n,O)
12	Compute temporal occupancy and energy imbalance versus beta.	Inertial branch
13	Apply parity, native charge/action, phase transport, and fine/speed closure.	Electromagnetic branch
14	Compare only completed outputs with independent witness packets.	Validation

The algorithm is bidirectional where the carrier and rendering relations are invertible. A downstream witness may be used to audit a completed path, but it does not replace the upstream Cell3 generator.

15. Cross-scale witness placement

The Solar-System, galaxy, molecular, atomic, magnetic, and metrology packets occupy downstream nodes of the same graph. They do not introduce new primitive geometry.

Witness family	Existing K-Field node	Reading
Solar-System dynamics	D07 reciprocal geometry; D08 Gamma2; D10-D11 gravity compiler	Macroscopic directional sampling of the reciprocal gravity branch
Galaxy distribution	Mixed primitive lifts, seven-sheet grammar, occupancy response	Macroscopic occupation of the transport grammar
Atomic lengths and spectra	D04-D05 fine/speed, reduced mass, action	Dimensional branch return
Magnetic moments	D06 with eta_N	Magneton ratio returns to Mass Bridge
Electrical metrology	D06 charge-action skeleton	Josephson, von Klitzing, and conductance closure
Materials and resonators	H(k), D, q-shells, polarity planes	Local phase-holonomy response

The use of a common graph means that each witness must preserve the same carrier ratios, reciprocal orientation, parity grammar, and rendering sequence. A witness is accepted only when its forward and reverse paths commute through the same kernel.

16. Foundational synthesis

The K-Field is generated by an exact primitive grammar rendered through an oblique Cell3 metric. The direct metric supplies rest geometry and dimensional closure. The face cofactors supply mass, fine/speed, and gravity carriers. The reciprocal metric supplies propagation and anisotropic convergence. The address sum supplies sheets and parity. The three primitive lifts supply the coherent phase-transfer mechanism.

Gravity, inertia, and electromagnetism are not independent primitive additions. They are typed branch readings of one Cell3 closure state. Gravity is the even Mass-Bridge occupancy branch propagated through reciprocal geometry and rendered through Gamma2, the dimensional bridge, and the directional compiler. Inertia is the rate-dependent mismatch among transit time, Compton occupancy, de Broglie phase, and kinetic energy across the fixed closure length. Electromagnetism is the odd polarity-phase branch rendered through native charge, action, fine/speed, medium, magnetic, and metrology relations.

Converged closure. Derived Cell3 geometry -> primitive faces and reciprocal geometry -> carrier triangle -> native unit grammar -> dimensional clocks and rates -> reciprocal transport -> gravity, inertia, and electromagnetism as even, rate, and odd branch readings. The branches differ in physical interpretation but return to the same geometry, ratios, and native units.

Appendix A. Complete Cell3 vertex ledger

Address	x (nm)	y (nm)	z (nm)	N dot p (nm)
000	0.000000000000000	0.000000000000000	-0.000000000000000	0.000000000000000
001	-0.010129317626665	-0.389787702363804	-0.689734408362922	0.302545376084970
010	0.035330775844051	0.207319850441498	-0.547401085391957	0.302545376084970
011	0.025201458217386	-0.182467851922307	-1.237135493754880	0.605090752169939
100	0.327501242069651	-0.054897007886313	-0.037487502896305	0.302545376084970
101	0.317371924442986	-0.444684710250118	-0.727221911259228	0.605090752169939
110	0.362832017913702	0.152422842555185	-0.58488588288262	0.605090752169939
111	0.352702700287037	-0.237364859808620	-1.274622996651185	0.907636128254909

Appendix B. Complete 27-node Lucas ledger

i	j	k	lambda	parity	x (nm)	y (nm)	z (nm)
-1	-1	-1	-3	-	-0.176351350144	0.118682429904	0.637311498326
-1	-1	0	-2	+	-0.181416008957	-0.076211421278	0.292444294144
-1	0	-1	-2	+	-0.158685962221	0.222342355125	0.363610955630
0	-1	-1	-2	+	-0.012600729109	0.091233925961	0.618567746877
-1	-1	1	-1	-	-0.186480667770	-0.271105272459	-0.052422910037
-1	0	0	-1	-	-0.163750621035	0.027448503943	0.018743751448
-1	1	-1	-1	-	-0.141020574299	0.326002280346	0.089910412934
0	-1	0	-1	-	-0.017665387922	-0.103659925221	0.273700542696
0	0	-1	-1	-	0.005064658813	0.194893851182	0.344867204181
1	-1	-1	-1	-	0.151149891926	0.063785422018	0.599823995429
-1	0	1	0	+	-0.168815279848	-0.167445347239	-0.326123452733
-1	1	0	0	+	-0.146085233113	0.131108429164	-0.254956791248
0	-1	1	0	+	-0.022730046735	-0.298553776403	-0.071166661485
0	0	0	0	+	0.000000000000	0.000000000000	0.000000000000
0	1	-1	0	+	0.022730046735	0.298553776403	0.071166661485
1	-1	0	0	+	0.146085233113	-0.131108429164	0.254956791248
1	0	-1	0	+	0.168815279848	0.167445347239	0.326123452733
-1	1	1	1	-	-0.151149891926	-0.063785422018	-0.599823995429
0	0	1	1	-	-0.005064658813	-0.194893851182	-0.344867204181
0	1	0	1	-	0.017665387922	0.103659925221	-0.273700542696
1	-1	1	1	-	0.141020574299	-0.326002280346	-0.089910412934
1	0	0	1	-	0.163750621035	-0.027448503943	-0.018743751448
1	1	-1	1	-	0.186480667770	0.271105272459	0.052422910037
0	1	1	2	+	0.012600729109	-0.091233925961	-0.618567746877
1	0	1	2	+	0.158685962221	-0.222342355125	-0.363610955630
1	1	0	2	+	0.181416008957	0.076211421278	-0.292444294144
1	1	1	3	-	0.176351350144	-0.118682429904	-0.637311498326

Appendix C. Matrix and identity ledger

C.1 Direct metric M

M row	a	b	c
a	0.111676057905435	0.020710333281606	0.043937235097789
b	0.020710333281606	0.343877732397120	0.296392758961523
c	0.043937235097789	0.296392758961523	0.627770610069386

C.2 Reciprocal metric R

R row	a*	b*	c*
a*	9.208025528683759	0.001534802428124	-0.645188035676901
b*	0.001534802428124	4.903398312306198	-2.315175584416389
c*	-0.645188035676901	-2.315175584416389	2.731171274746107

C.3 Weighted branch Laplacian L_w

L_w row	a	b	c
a	166.983836268130	-0.396285985688	-166.587550282441
b	-0.396285985688	598.174619965640	-597.778333979952
c	-166.587550282441	-597.778333979952	764.365884262393

C.4 Machine-precision identities

Identity	Computed residual
$N \cdot (a/2) = N \cdot (b/2) = N \cdot (c/2) = h$	2.776e-17
$s_{ac} - s_{ab} - s_{bc} = 0$	3.469e-18
$L_w [1,1,1]^T = 0$	0.000e+00
$G_{base} - \text{Gamma2 } C_{G,pred} = 0$	0.000e+00
$T_G/\tau_e - S_{CMB} = 0$	0 by definition
$\rho^2 \text{ quotient-fiber coupling}$	0.543701653538894

Appendix D. Symbol ledger

Symbol	Definition
B	Ordered Cell3 basis with columns a, b, c
M	Direct metric $B^T B$
R	Reciprocal metric M^{-1}
Rhat	Normalized reciprocal metric $R/qbar_F$
L_G	Cell3 closure length $ a+b+c $
T_G	Light-crossing time L_G/c
tau_R	One native closure update
tau_e	Electron Compton period
v_F	Fine-leg velocity c/c_N
L_N	Fine-leg length $\hbar/(m_e v_F)$
tau_N	Fine-leg time L_N/v_F
A_ab,A_ac,A_bc	Primitive native face areas
eta_N	Mass Bridge 1000 A_bc/A_ab
alpha_N	Fine factor $A_ac/(100 A_bc)$
c_N	Inverse fine/speed factor $1/\alpha_N$
Gamma2	Native gravitational scalar
Rhat	Normalized reciprocal response metric
r_K	Native reciprocal-metric distance
L_K	$-\text{div}_K(Rhat \text{ grad}_K)$
N_m	Native matter occupancy
Q_m	Mass-polarization charge $\eta_N N_m$
Phi_m	Native source potential
H(k)	Three-channel phase-transfer factor
chi	Address parity $(-1)^{(i+j+k)}$
q0,N	Native unsigned charge magnitude
G_base	Gamma2 $C_G, pred$
G_eff	Observer-direction rendered gravity

Appendix E. Corpus source ledger

The following generated corpus references supply the formulas, numerical ledgers, branch definitions, and dependency order used in this reconstruction.

Artifact	Primary role
K-Field framework restated	Cell3 as oblique rendering of the centered three-state primitive grammar
K-Field Typed Graph Bundle	Cell3 geometry, laminar transfer operator, inertial phase spaces, and branch graph
K-Field Temporal Phase Imbalance	Locked-cell inertia, native clocks, phase counts, and energy surplus
K-Field Hidden New Pathways	D00-D14 dependency architecture and gravity-recursion branch
The Twelve Locks	Carrier triangle, Mass Bridge, fine/speed, native units, magneton return, and EM/metrology closure
K-Field Gravity Is Not a Primitive	Mass occupancy source, reciprocal field operator, inverse-distance potential, and compiler collapse
K-Field Gravity: Least-Cost Convergence of Mass-Polarized Traversal	Rendered gravitational parameter and equivalence cancellation
Cell3 G Surface	Directional gravity surface and phase metric
K-Field Solar-System Witness	Containment of planetary observations within existing D00-D14 nodes
K-Field Bridge to the Top 100 Physics Formulas	Shared generative grammar and cross-domain derivative operations